Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

Speed, Distance and Time

Unit 2: Forces and Motion • the formula triangle

Two measurements, one word

Every speed comes from exactly two measurements: **how far** something went, and **how long** it took. Put them together and you have its speed. You have already measured your own walking and running speeds; today you practise the formula behind them.

The formula

$$\textbf{speed} \;=\; \boxed{v = \frac{s}{t}} \;=\; \frac{\textbf{distance}}{\textbf{time}}$$

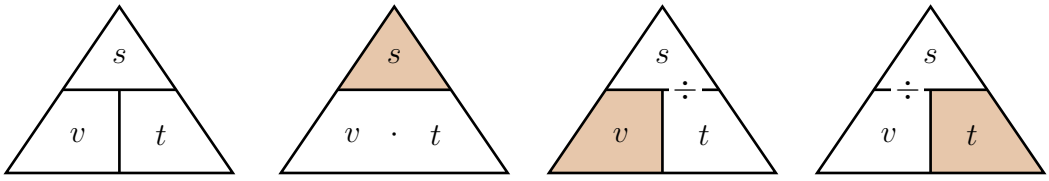
The line just means divide. $\frac{s}{t}$ and $s \div t$ are the same instruction written two ways.

Physicists write **distance as** s and **speed as** v. These are the letters you will use all the way to GCSE.

Quantity	Letter	Unit
distance	s	metres (m)
speed	v	metres per second (m/s)
time	t	seconds (s)

Triangle math

Put the letters in the triangle. **Shade the one you are looking for**, and what is left in the other two boxes tells you what to do: **one above the other means divide**, **side by side means multiply**.



the triangle **shade** s**:** $s = v \cdot t$ **shade** v**:** $v = s \div t$ **shade** t**:** $t = s \div v$

The top of the triangle, s**, is distance.** The two at the bottom, speed and time, multiply together to give it.

Worked examples

One for each corner of the triangle. Every answer follows the same four steps.

Worked example — copy this shape

A girl walks **20 m** in **10 s**. What is her speed?

WRITE $s = 20$ m $t = 10$ s $v = ?$
CHOOSE shade v: $v = s \div t$
PUT IN $v = 20 \div 10$
ANSWER $v =$ **2 m/s**

Worked example — copy this shape

A runner runs at **4 m/s** for **25 s**. How far does she run?

WRITE $v = 4$ m/s $t = 25$ s $s = ?$
CHOOSE shade s: $s = v \cdot t$
PUT IN $s = 4 \cdot 25$
ANSWER $s =$ **100 m**

Worked example — copy this shape

A dog runs **60 m** at **6 m/s**. How long does it take?

WRITE $s = 60$ m $v = 6$ m/s $t = ?$
CHOOSE shade t: $t = s \div v$
PUT IN $t = 60 \div 6$
ANSWER $t =$ **10 s**

Check: every answer ends with its unit — **m**, **s** or **m/s** — matching the letter you were looking for in the table above.

Your turn

Use the four steps for every question. Shade your own triangle in the margin if it helps.

Find the speed

(1) A boy walks **30 m** in **15 s**. What is his speed?

WRITE $s = 30$ m $t = 15$ s $v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 30 \div 15$

ANSWER $v = 2$ **m/s**

(2) A cyclist rides **150 m** in **30 s**. What is her speed?

WRITE $s = 150$ m $t = 30$ s $v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 150 \div 30$

ANSWER $v = 5$ **m/s**

(3) A car drives **400 m** in **20 s**. What is its speed?

WRITE $s = 400$ m $t = 20$ s $v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 400 \div 20$

ANSWER $v = 20$ **m/s**

Find the distance

(4) A horse trots at **5 m/s** for **12 s**. How far does it go?

WRITE $v = 5$ m/s $t = 12$ s $s = ?$

CHOOSE $s = v \cdot t$

PUT IN $s = 5 \cdot 12$

ANSWER $s = 60$ **m**

(5) A sailing boat moves at **3 m/s** for **60 s**. How far does it sail?

WRITE $v = 3$ m/s $t = 60$ s $s = ?$

CHOOSE $s = v \cdot t$

PUT IN $s = 3 \cdot 60$

ANSWER $s = 180$ **m**

(6) A train travels at **50 m/s** for **8 s**. How far does it travel?

WRITE $v = 50$ m/s $t = 8$ s $s = ?$

CHOOSE $s = v \cdot t$

PUT IN $s = 50 \cdot 8$

ANSWER $s = 400$ **m**

Find the time

(7) A swimmer swims **50 m** at **2 m/s**. How long does it take?

WRITE $s = 50$ m $v = 2$ m/s $t = ?$

CHOOSE $t = s \div v$

PUT IN $t = 50 \div 2$

ANSWER $t = 25$ **s**

(8) A cheetah sprints **120 m** at **30 m/s**. How long does it take?

WRITE $s = 120$ m $v = 30$ m/s $t = ?$

CHOOSE $t = s \div v$

PUT IN $t = 120 \div 30$

ANSWER $t = 4$ **s**

(9) A plane flies **2400 m** at **200 m/s**. How long does it take?

WRITE $s = 2400$ m $v = 200$ m/s $t = ?$

CHOOSE $t = s \div v$

PUT IN $t = 2400 \div 200$

ANSWER $t = 12$ **s**

To go further

(a) A robot drives **3 m** in **12 s**. What is its speed?

$v = s \div t = 3 \div 12 =$ **0.25** *m/s — a quarter of a metre every second.*

(b) Usain Bolt ran **100 m** in about **10 s**. A snail moves at about **0.01 m/s**. How long would the snail take to cover the same 100 m?

$t = s \div v = 100 \div 0.01 =$ **10 000** *s — nearly three hours, to Bolt's ten seconds.*